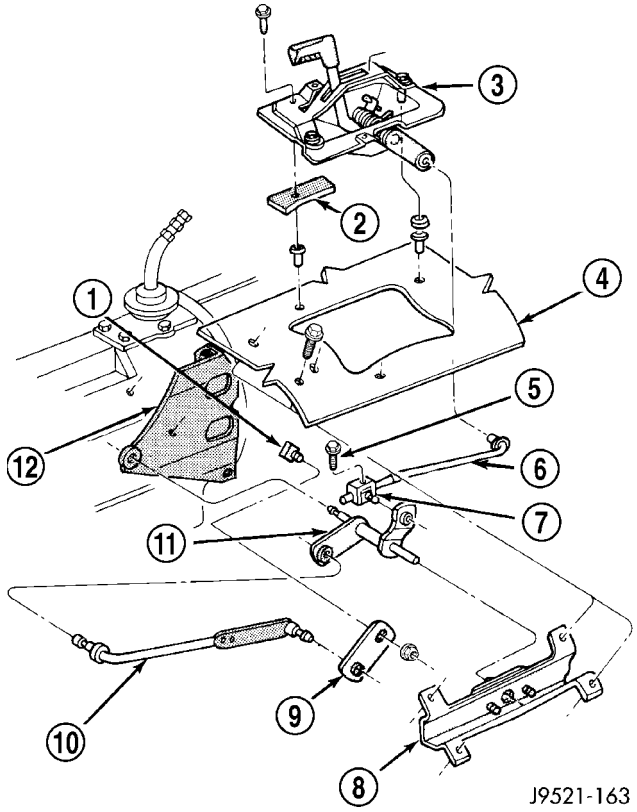

Fig. 103 Shift Linkage

1 - TRANSFER CASE SHIFT LEVER SHAFT
2 - SEAL
3 - TRANSFER CASE SHIFT LEVER ASSEMBLY
4 - FLOORPAN
5 - TRUNNION LOCK BOLT
6 - SHIFT ROD
7 - ADJUSTING TRUNNION
8 - TORQUE SHAFT BRACKET
9 - RANGE LEVER
10 - TORQUE SHAFT ROD
11 - TORQUE SHAFT
12 - LINKAGE BRACKET

VEHICLE SPEED SENSOR

DESCRIPTION

The 3-wire Vehicle Speed Sensor (VSS) (Fig. 104) is located in the transfer case rear extension.

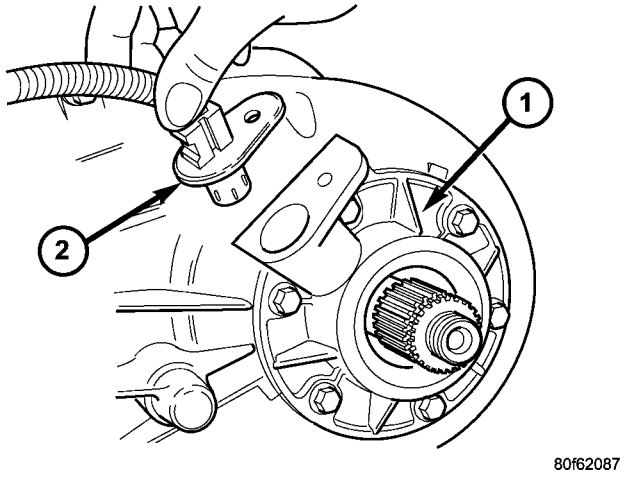

Fig. 104 Speedometer Sensor

1 - EXTENSION HOUSING
2 - SPEEDOMETER SENSOR

OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

Under deceleration conditions, the PCM adjusts the Idle Air Control (IAC) motor to maintain a desired MAP value. Under idle conditions, the PCM adjusts the IAC motor to maintain a desired engine speed.

REMOVAL

The Vehicle Speed Sensor (VSS) is located in the transfer case rear extension.

(1) Raise and support vehicle.

(2) Disconnect electrical connector from sensor by pushing slide tab. After slide tab has been positioned, push in on secondary release lock on side of connector and pull connector from sensor.

(3) Remove sensor mounting bolt.

(4) Remove sensor (pull straight out) from rear extension.